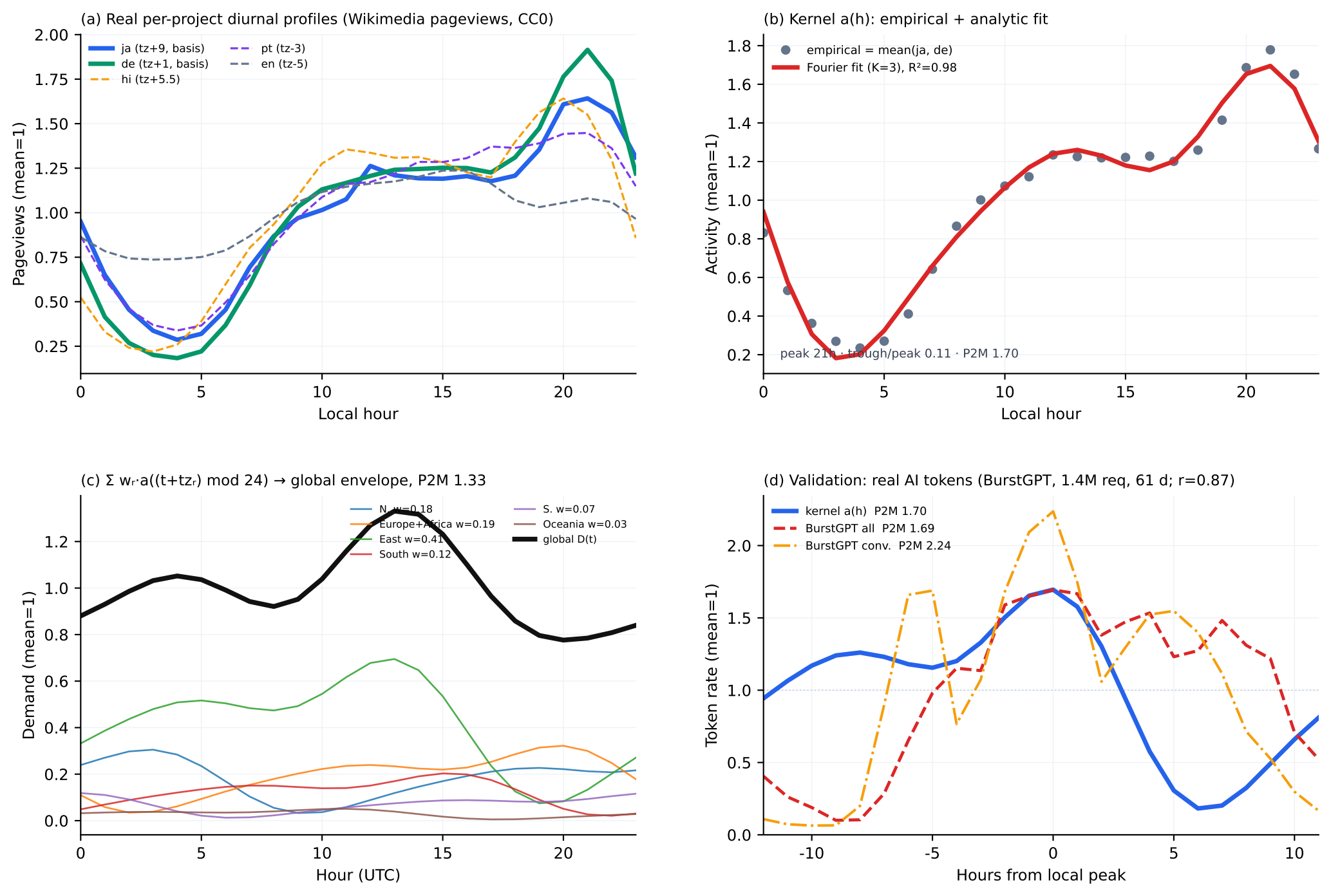
(a) Real per-project diurnal profiles (Wikimedia pageviews, CC0)
ja (tz+9, basis)
de (tz+1, basis)
hi (tz+5.5)
pt (tz-3)
en (tz-5)
Pageviews (mean=1)
Local hour
(b) Kernel a(h): empirical + analytic fit
empirical = mean(ja, de)
Fourier fit (K=3), R²=0.98
peak 21h · trough/peak 0.11 · P2M 1.70
Activity (mean=1)
Local hour
(c) Σ wᵣ·a((t+tzᵣ) mod 24) → global envelope, P2M 1.33
N. w=0.18
Europe+Africa w=0.19
East w=0.41
South w=0.12
S. w=0.07
Oceania w=0.03
global D(t)
Demand (mean=1)
Hour (UTC)
(d) Validation: real AI tokens (BurstGPT, 1.4M req, 61 d; r=0.87)
kernel a(h) P2M 1.70
BurstGPT all P2M 1.69
BurstGPT conv. P2M 2.24
Token rate (mean=1)
Hours from local peak